Harmonic Motion Practicum

Experiment to Determine Gravitational Acceleration Using a Pendulum and Spring Oscillation

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I. Objective

To determine the value of gravitational acceleration from harmonic oscillations in a simple pendulum and spring motion. For the pendulum experiment the formula used is:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

For the spring experiment the formula used is:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

II. Theoretical Background

A. Harmonic Motion

Simple harmonic motion (SHM) is a type of periodic motion where the restoring force acting on an object is proportional to the displacement from its equilibrium position and directed toward that equilibrium. The general equation describing this motion is

$$F = -kx$$

where
F = restoring force
k = force constant
x = displacement from equilibrium
This type of motion produces oscillations with a constant period.

B. Simple Pendulum

A simple pendulum consists of a small mass (called the bob) suspended from a fixed point by a light, inextensible string. When displaced slightly from its equilibrium position and released, the pendulum oscillates back and forth due to gravity. The restoring force acting on the pendulum comes from the component of gravitational force along the arc of motion. For small angles:

$$F = -mg\sin\theta$$

For small angles (θ in radians):

$$\sin\theta \approx \theta$$

Thus the equation becomes:

$$F = -mg\theta$$

Because the arc displacement $x = l\theta$, the motion can be shown to follow simple harmonic motion. From the SHM equation, the period of oscillation is obtained:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where

T = period of oscillation (s)
l = length of the string (m)
g = gravitational acceleration (m/s²)

By squaring both sides:

$$T^2 = 4\pi^2\frac{l}{g}$$

This equation shows that T^2 is proportional to the length of the pendulum. Therefore, a graph of T^2 versus l l will produce a straight line whose slope equals

C. Spring Oscillation

A spring that obeys Hooke's Law experiences a restoring force proportional to its displacement:

$$F = -kx$$

where

F = restoring force
k = force constant
x = displacement from equilibrium

When a mass m is attached to a vertical spring and displaced from its equilibrium position, the system oscillates in simple harmonic motion.

Using Newton's Second Law:

$$F = ma$$

Substituting Hooke's Law:

$$ma = -kx$$

or:

$$a = -\frac{k}{m}x$$

This matches the differential equation of simple harmonic motion. From this equation, the oscillation period is

$$T = 2\pi\sqrt{\frac{m}{k}}$$

However, in a vertical spring system, the equilibrium extension of the spring is determined by the balance of gravitational force and spring force:

$$mg = k\Delta l$$

Which gives

$$k = \frac{mg}{\Delta l}$$

Substituting this into the period equation:

$$T = 2\pi\sqrt{\frac{m}{mg/\Delta l}}$$

Which simplifies to

$$T = 2\pi\sqrt{\frac{\Delta l}{g}}$$

Squaring both sides:

$$T^2 = 4\pi^2\frac{\Delta l}{g}$$

This equation shows that T^2 is proportional to the extension of the spring. Therefore, a graph of T^2 versus Δl will produce a straight line, and from the slope we can determine the value of gravitational acceleration.

III. Tools and Materials

A. Simple Pendulum

1. Retort stand (support stand) – 1 piece
2. Clamp – 1 piece
3. String – as needed
4. 30 cm ruler – 1 piece
5. Scissors – 1 piece
6. Stopwatch – 1 piece
7. Metal mass (pendulum bob) – 1 piece

B. Spring

1. Retort stand (support stand) – 1 piece
2. Clamp – 1 piece
3. String – as needed
4. Spiral spring – 1 piece
5. Scissors – 1 piece
6. Stopwatch – 1 piece
7. 30 cm ruler – 1 piece
8. Metal masses (weights) – 5 pieces

IV. Procedure

A. Pendulum Oscillation Experiment

1. Prepare the required tools and materials.
2. Set up the stand and attach the clamp to the upper part of the stand.
3. Adjust the screw on the clamp until the desired height and a straight angle are obtained.
4. Prepare a data table on a sheet of paper.
5. Measure a string with a certain length (40 cm) using a 30 cm ruler and a 15 cm ruler for the remaining part, then cut the string.
6. Tie one end of the string to the 50 g metal pendulum bob and the other end to the clamp. When tied, the string length may decrease, so measure the string again after the pendulum is suspended and record the length lll.
7. Pull the pendulum to an angle of about 10–15° while preparing the stopwatch.
8. Start the stopwatch when the pendulum is released.
9. Count 10 oscillations of the pendulum (one oscillation is counted when the pendulum moves from the starting point to the opposite side and returns to the starting point).
10. Stop the stopwatch.

11. Record the measured time.
12. Repeat steps 6–10 three times for the same string length.
13. Vary the string length five times by adjusting the clamp, then repeat steps 5–10.
14. Calculate the average time.
15. Calculate the average time for each trial (since the results in step 12 may differ).
16. Calculate the period (T) using the formula

$$T = \frac{t}{n}$$

Where $n = 10$ oscillations.

17. Calculate the square of the period T^2.
18. Convert all values into International units.

19. Plot a graph with period squared (T^2) on the Y-axis and string length (l) on the X-axis.
20. Plot the points based on the data and connect them to form a piecewise line representing the data trend.
21. Draw a new straight line (the gradient line) that evenly divides the area formed by the graph in step 20.
22. Select two arbitrary points from this gradient line (not from the plotted data points) and record them.
23. Take the string length values from the two points, subtract them, and record the difference

$$\Delta l = l_2 - l_1$$

24. Take the squared period values from the two points, subtract them, and record the difference

$$\Delta T^2 = T^2_2 - T^2_1$$

25. Substitute these values into the formula to determine the gravitational acceleration (g).

B. Spring Experiment

1. Prepare the tools and materials for the second experiment, especially the spiral spring and a mass.
2. Set up the stand and attach the clamp to the upper part of the stand.
3. Measure the initial length of the spring before adding a load using a 30 cm ruler and record it as l_0.
4. Tie the spiral spring to the clamp using a small piece of string.
5. Secure the spring to the clamp by tying the string twice.
6. Hang a mass mmm at the end of the spiral spring.

7. Measure the length of the spring after the mass is attached using the ruler.
8. Pull the spring slightly downward.
9. Start the stopwatch when the spring is released and measure the time required for 10 oscillations.
10. Repeat steps 5–8 five to six times, each time increasing the mass by adding additional weights, and measure the spring length again.
11. Record the final spring length, the extension of the spring, and the oscillation time in the data table.

V. Data and/or Data Tables

A. Pendulum Oscillation Experiment

No.	t_1 (s)	t_2 (s)	t_3 (s)	Average t (s)	T (s)	T^2 (s^2)	l (m)
1	11	11	11	11	1.10	1.21	0.275
2	10.8	10.2	10.8	10.6	1.06	1.12	0.245
3	9.2	9	9	9.06	0.906	0.82	0.176
4	7.2	7.8	7.4	7.46	0.746	0.55	0.106
5	6.2	6.2	6.2	6.2	0.62	0.38	0.074

B. Spring Experiment

:

No.	m (kg)	l (m)	Δl (m)	t_1 (s)	t_2 (s)	t_3 (s)	Average t (s)	T (s)	T^2 (s^2)
1	0.05	0.182	0.039	4.4	4	4.6	4.33	0.43	0.18
2	0.07	0.20	0.057	5	4.8	5	4.93	0.493	0.24
3	0.09	0.22	0.077	5.8	5.6	5.8	5.73	0.573	0.328
4	0.10	0.229	0.086	6	5.6	6	5.86	0.585	0.34
5	0.12	0.245	0.102	6.4	6.5	6.4	6.43	0.643	0.41

For the calculations:
The following formulas are used:

$$T = \frac{t}{n}$$

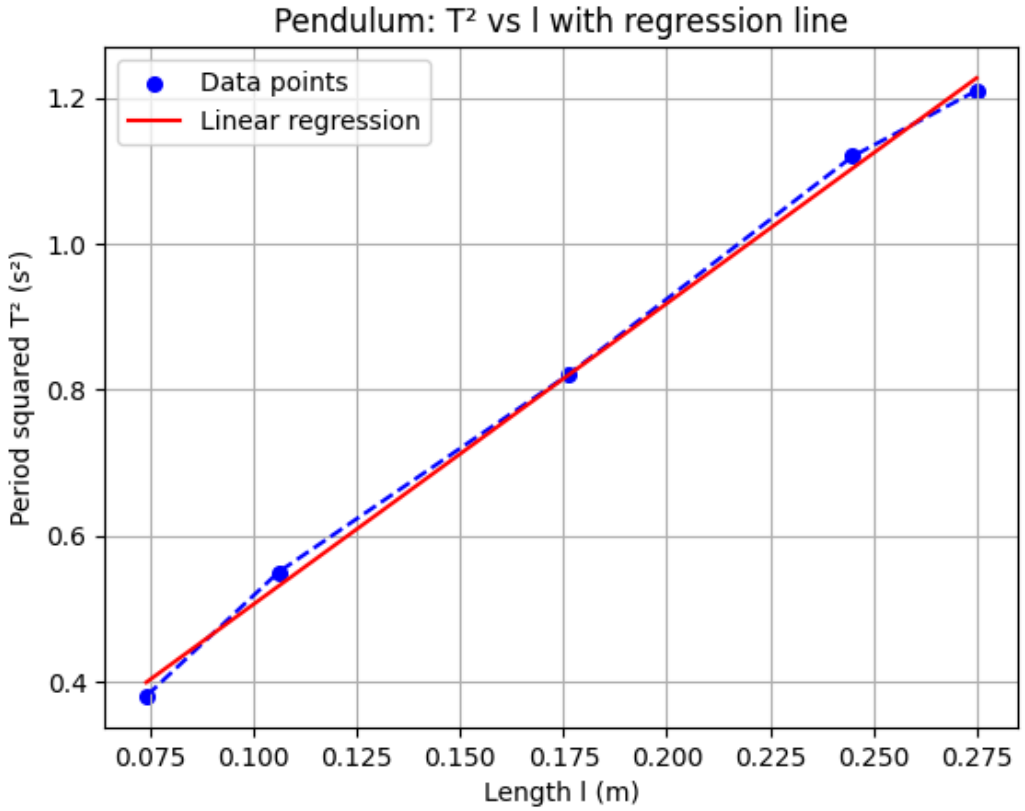
$$\Delta l = l - l_0$$

Where
n = 10 indicates 10 oscillations,
l_0 = 0.143 is the initial length of the spring

VI. Data Analysis and Discussion

A. Pendulum

The resulting graph is as follows:



From the graph, the following equation is obtained:

$$T^2 = 4.121 * l + 0.094$$

To test the sample using two points, the following data were obtained:

Rope Length (l) (m)	Period Squared (T^2) (s)
0.1	0.5060771260564986
0.2	0.9182086073643466

With:

$$\Delta l = l_2 - l_1$$
$$\Delta T^2 = T^2_2 - T^2_1$$

The gravitational acceleration can then be calculated using the formula:

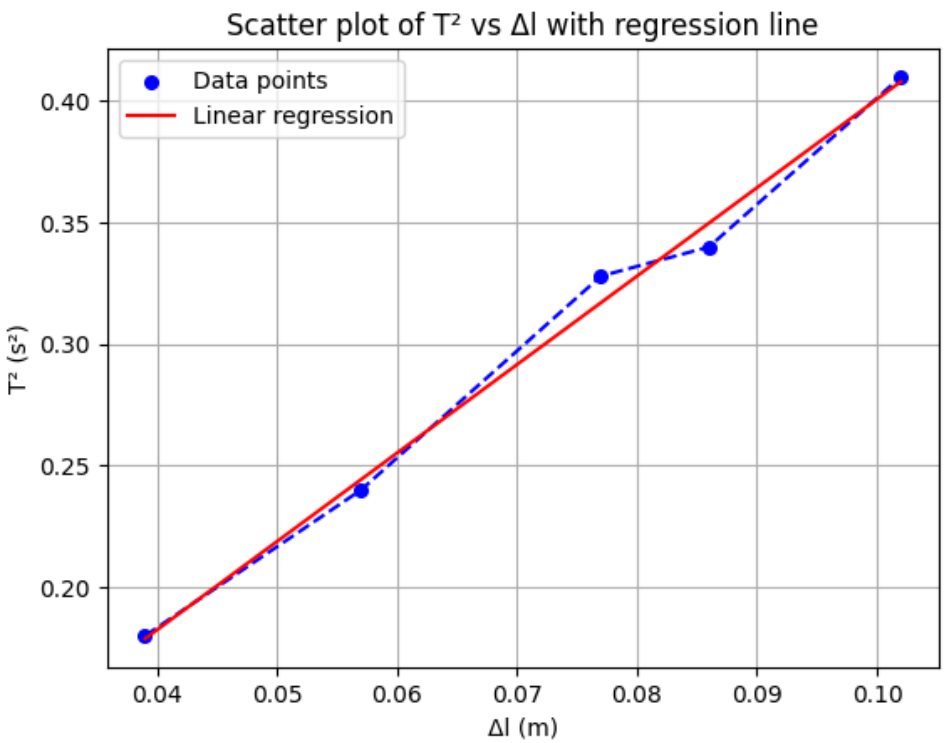
$$g = 4\pi^2 \frac{\Delta l}{\Delta T^2}$$

Substituting the values:

$$g = 4\pi^2 \frac{0{,}2-0{,}1}{0.9182086073643466-0.5060771260564986}$$
$$g = 4\pi^2 \frac{0{,}1}{0.41213148130784805} \approx 9.58\, m/s^2$$
$$g \approx 9.58\, m/s^2$$

B. Spring

The resulting graph is as follows:



From the graph, the equation obtained is:

$$T^2 = 3.639 * \Delta l + 0.037$$

Using two sample points from the gradient:

Rope Length (l) (m)	Period squared (T^2) (second)
0.05	0.21881271562345983
0.06	0.255203384261541

With

$$\Delta l = \Delta l_2 - \Delta l_1$$
$$T^2 = T^2_2 - T^2_1$$

Then:

$$g = 4\pi^2 \frac{\Delta l}{T^2}$$

Substituting the values:

$$g = 4\pi^2 \frac{0{,}06-0{,}05}{0.255203384261541 - 0.21881271562345983}$$
$$g = 4\pi^2 \frac{0{,}01}{0.03639066863808116} \approx 10,85\ m/s^2$$
$$g \approx 10,85\ m/s^2$$

In the experiment, one of the steps was drawing a graph and determining its gradient. In this context, the gradient represents the average relationship between the squared period and the string length in the pendulum experiment, and the relationship between the squared period and the extension of the spring in the spring experiment.

After drawing the gradient line, two arbitrary points are selected from this line. The differences between these points are calculated and substituted into the equation to determine the gravitational acceleration. However, the obtained value is not always exact.

In the first experiment, calculating the differences is necessary because when the string length decreases, the oscillation time and period also decrease. If the values are directly substituted into the formula

$$g = 4\pi^2 \frac{l}{T^2},$$

the calculated gravitational acceleration would vary depending on the string length. Therefore, calculating the differences between two points helps obtain a more consistent value of gravitational acceleration. In both experiments, the calculated values of gravitational acceleration range between 9 m/s² and 10 m/s².

If we examine the first experiment again, the independent variable is the length of the string (l). By changing the string length, the oscillation time ttt changes

accordingly. When the string length becomes shorter, the oscillation time and period also become shorter. This observation was clearly visible during the experiment: the shorter the string, the faster the pendulum oscillates.

In the second experiment, the independent variable is the mass of the load. By varying the mass, the extension of the spring changes, which affects the oscillation period. When the extension of the spring is smaller (meaning the load mass is smaller), the time required for one oscillation is also shorter.

The standard value of gravitational acceleration is 9.8 m/s².
In the spring experiment, when $\Delta l = 0.05\ m$, the formula predicts:

$$T^2 = 4\pi^2 \frac{\Delta l}{g}$$

Substituting the values:

$$T^2 = 4\pi^2 \frac{0.05}{9.8} \approx 0.0026\ s^2$$

However, the data obtained from the graph shows that:

$$T^2 = 0.2188\ s^2$$

This value was taken from the average line of the graph. According to the theoretical equation, the difference in period should be smaller, which means the line should be steeper rather than more horizontal.

To produce a steeper line, some of the experimental values such as 0.34 s² (for Δl = 0.086 m) and 0.41 s² (for Δl = 0.102 m) could theoretically be slightly larger if the measurements were more precise. Since the period formula is

$$T = \frac{t}{n}$$

the measured time *t* might be slightly higher in more precise conditions. The reason why the experimental gravitational acceleration does not exactly equal 9.8 m/s² is due to several sources of experimental error. One major factor is the presence of air currents, which affect the motion of the pendulum when it swings freely. The theoretical formulas assume that the experiment takes place in a vacuum without air resistance. Other factors include measurement inaccuracies, reaction time when using the stopwatch, and limitations of the measuring instruments.

VII. Conclusion

Based on the experiment and the group's analysis, it can be concluded that experiments involving Newton's Law of Gravitation and Newton's Second Law can be used to determine the value of gravitational acceleration using a simple pendulum

experiment and a spring force experiment. Through the simple pendulum experiment with varying string lengths and the spring experiment with different load masses, we were able to determine the value of gravitational acceleration based on these physical laws. The results obtained varied slightly. Both the length of the pendulum string and the mass of the load on the spring show a direct relationship with the resulting period, although the differences are relatively small. It should be noted that these variations may also be caused by several factors related to the precision of the experimental procedure. These factors include environmental conditions and the condition or limitations of the equipment used during the experiment. Although the calculated gravitational acceleration obtained by the group may differ from the standard gravitational value, the formulas used to determine gravitational acceleration were still proven valid. Therefore, future researchers can still use these formulas as a method to determine gravitational acceleration.

VIII. References

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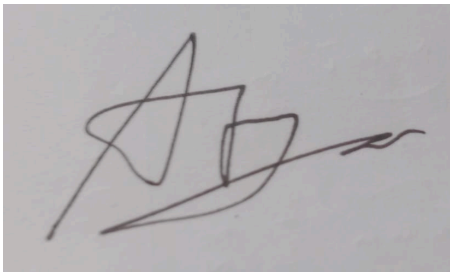

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